

Holcim: 84 plants and the carbon bill behind them

84 register assets worth about \$22.2bn. Expected cost to 2050 runs from \$4.2bn under current policies to \$22.7bn under Net Zero 2050, and the range between scenarios is the transition story [1].

ClimRisk risk analyst · 2026-10-07



JAS 2026. The Oceanic Niño Index reads +2.16 °C: El Niño. Since 1950 it has crossed +2 °C only 4 times [2]. The ocean is the loud part of the climate story this year. What follows is about the part that does not make the news: what climate policy and climate hazards do to the value of a specific set of industrial assets.

Holcim in six numbers

Transition and physical cost to 2050, portfolio level

84

register assets in the portfolio

Climate TRACE v6 + EPA GHGRP

\$22.2bn

estimated portfolio value

capacity-based proxy unless supplied

\$4.2bn

expected cost, current policies

Monte Carlo mean, PV 2026–2050

\$14.0bn

expected cost, delayed transition

Monte Carlo mean, PV 2026–2050

\$22.7bn

expected cost, Net Zero 2050

Monte Carlo mean, PV 2026–2050

\$16.8bn

95th percentile, delayed transition

1-in-20 outcome

Source: ClimRisk risk analyst; NGFS Phase 5; asset register

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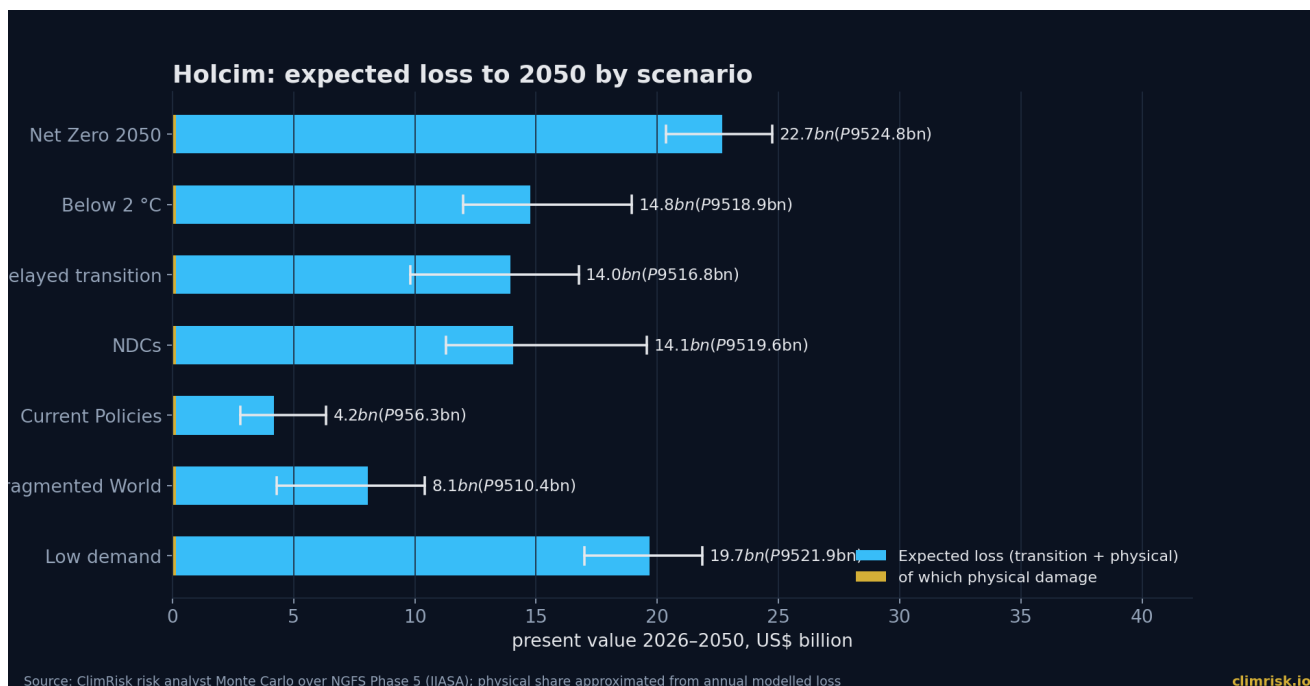
Portfolio numbers from the engine's Monte Carlo over NGFS carbon-price and activity paths.

90× quiet to loud

The quiet risk is about 90 times the loud one: \$13.8bn of expected cost is transition against about \$153.5m physical, in the delayed transition. Physical is approximated as each asset's modelled annual loss over 25 years at 7%.

The loud risk: the scenario spread

The headline for transition risk is a range, not a point. Across the seven NGFS scenarios the portfolio's expected cost to 2050 runs from \$4.2bn to \$22.7bn; Net Zero 2050 costs 5.4 times current policies [1]. Which scenario the plan sits in is the first question for any lender.



Scenario	Mean	Median	95th pct	Expected shortfall
Net Zero 2050	\$22.7bn	\$22.9bn	\$24.8bn	\$25.0bn
Below 2°C	\$14.8bn	\$13.3bn	\$18.9bn	\$19.3bn
Delayed transition	\$14.0bn	\$15.4bn	\$16.8bn	\$17.0bn
Nationally Determined Contributions (NDCs)	\$14.1bn	\$12.2bn	\$19.6bn	\$20.0bn
Current Policies	\$4.2bn	\$3.7bn	\$6.3bn	\$6.5bn
Fragmented World	\$8.1bn	\$9.5bn	\$10.4bn	\$10.6bn
Low demand	\$19.7bn	\$20.1bn	\$21.9bn	\$22.2bn

The quiet risk: which assets carry it

Where the emissions sit decides which policies reach the site. The portfolio number hides concentration: the three largest assets carry 14% of the delayed-transition cost.



The ten assets with the largest expected delayed-transition cost [3].

Asset	Country	Sector	Emissions	Carbon price now	Delayed-transition cost
HOLCIM (US) INC STE GENEVIEVE PLANT	USA	ghgrp-327	2.97 Mt	\$0/t	\$889.5m
HOLCIM (US) INC. D/B/A LAFARGE ALPEN A	USA	ghgrp-327	1.77 Mt	\$0/t	\$536.2m
Holcim (Argentina) Santa María Cement Plant	ARG	cement	2.00 Mt	\$4/t	\$503.1m
Amrize Alpena Cement Plant	USA	cement	1.54 Mt	\$0/t	\$410.6m
HOLCIM (US) Inc	USA	ghgrp-327	1.25 Mt	\$0/t	\$374.1m
HOLCIM (US) INC.	USA	ghgrp-327	1.21 Mt	\$0/t	\$364.4m
Holcim España Sagunto Cement Plant	ESP	cement	0.92 Mt	\$85/t	\$351.9m
Holcim (Belgique) Mons Cement Plant	BEL	cement	1.06 Mt	\$85/t	\$334.0m
Lafarge Ravena Cement Plant	USA	ghgrp-327	1.12 Mt	\$0/t	\$333.5m
Amrize Bloomsdale Cement Plant	USA	cement	0.95 Mt	\$0/t	\$331.7m

What this means, by reader

For the company. The lever is which assets abate first; the chart above is the order to ask about.

For the lender or investor. The exposure is a range. Ask which scenario the plan assumes, then read the matching row and its 95th percentile.

Method and limitations

Each asset run through the four-stage analyst; flood depths fetched live for the top 15 assets by emissions (others: cyclone + policy + transition only). Transition model draws are shared across assets (common scenario uncertainty); physical events independent. % of value is taken per draw against that draw's portfolio value, so it cannot exceed 100%. Asset values are estimated from capacity unless supplied, so replace them for a decision-grade figure. Direct costs only; second-round effects are not included. Where the expected cost approaches the estimated value, read it as the asset not being worth running at that price, and note that the value basis is a proxy with a wide band.

How ClimRisk measured this

- Register → every asset snapped to Climate TRACE / EPA GHGRP with emissions and capacity.
- Policy → World Bank carbon-pricing instruments by jurisdiction and sector.
- Scenarios → NGFS Phase 5 carbon price and output paths by region and model.
- Monte Carlo → shared draws across assets, 7 scenarios, discounted at 7%.
- Output → expected cost, percentiles and expected shortfall, asset by asset.

Sources

1. NGFS Phase 5 climate scenarios (IIASA Scenario Explorer) — carbon price and sector output, 7 scenarios, 3 models
<https://data.ene.iiasa.ac.at/ngfs/>
2. NOAA Climate Prediction Center, Oceanic Niño Index (ERSSTv5)
https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php
3. ClimRisk asset register — Climate TRACE v6 + EPA GHGRP 2023 with IBTrACS cyclone and terrain-elevation screening

Produced by the ClimRisk engine from the sources listed. Model outputs with the limitations stated; not forecasts or investment advice.